Finite-Time Synchronization of Fractional-Order Uncertain Quaternion-Valued Neural Networks via Sliding Mode Control

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Abstract

This work designs a fractional sliding mode control strategy to achieve finite-time synchronization for uncertain quaternion-valued neural networks. An appropriate sliding surface in quaternion space is constructed, and reaching conditions are proved ensuring trajectories reach the sliding surface within a predetermined finite settling time T_reach.

1. Introduction

While asymptotic synchronization guarantees convergence as time tends to infinity, practical engineering systems demand synchronization within a finite horizon. Finite-time synchronization possesses superior properties including faster response speed, enhanced disturbance rejection, and high precision.

Sliding Mode Control (SMC) is widely recognized as a robust nonlinear control technique against bounded external disturbances and model uncertainties. When combined with fractional calculus, fractional-order sliding surfaces provide additional degrees of freedom.

2. Preliminaries and Sliding Surface Design

We consider Caputo fractional derivative ^C D^alpha_t of order 0 < alpha < 1. Let e(t) = y(t) - x(t) in H^n be the synchronization error between drive and response neural networks.

Definition 2. *1 (Finite-Time Stability). The error system is said to achieve finite-time synchronization if lim_{t -> T*} ||e(t)|| = 0 and ||e(t)|| = 0 for all t >= T*, where T* = T(e(0)) < infinity is the finite settling time.*

Lemma 2. *4 (Fractional Finite-Time Inequality). Suppose a continuous function V(t) >= 0 satisfies ^C D^alpha_t V(t) <= -lambda V^gamma(t) for t >= 0, where lambda > 0, 0 < gamma < 1, and 0 < alpha < 1. Then V(t) reaches zero in finite time T_settle bounded by T_settle <= [V(0)^{1-gamma} Gamma(1 + alpha) / (lambda Gamma(1 - gamma + alpha))]^{1/alpha}.*

Assumption 3. *1 (Bounded External Disturbances). The unknown external disturbance d_i(t) in H satisfies ||d_i(t)|| <= D_i for all t >= 0, where D_i is a known positive constant.*

3. Sliding Mode Controller Synthesis

We design the fractional integral sliding surface as: s(t) = ^C D^{alpha-1}_t e(t) + int_0^t [K_1 e(tau) + K_2 sig^p(e(tau))] dtau, where sig^p(e) = |e|^p sign(e) with 0 < p < 1.

Theorem 3. *1 (Reachability of Sliding Surface in Finite Time). Under Assumption 3.1, with the sliding mode controller u(t) = u_{eq}(t) - (eta + D) sign(s(t)) - k s(t), the error trajectory reaches the sliding manifold s(t) = 0 in a finite time T_reach <= (||s(0)|| / eta)^{1/alpha}.*

Theorem 3. *2 (Finite-Time Synchronization on the Sliding Manifold). Once the error trajectory reaches the sliding manifold s(t) = 0, the error dynamics satisfies the fractional differential inequality of Lemma 2.4 and converges to zero within finite time T_sync = T_reach + T_settle.*

Remark 3. *6. The proposed fractional sliding surface eliminates chattering effects while ensuring invariant sliding motion against matched parameter uncertainties and non-differentiable disturbances.*